2.4 Modeling with Quadratic Functions

Name _____ Period ____ Date _____

Algebra II 2.4 (30-minute plan) · Student sheet

I can…

- write quadratic equations using vertex form and intercept form.
- use finite differences to tell whether data are linear, quadratic, or neither.
- model real-life data and situations with quadratic functions.

Vocabulary · fill in the missing word

Vertex form — $y = a(x - h)^2 + k$: use when you _____ the vertex and one more point

Intercept form — $y = a(x - p)(x - q)$: use when you know the _____ and one more point

First differences — the differences between _____ y-values (equally spaced x)

Second differences — the differences of the first differences; constant second differences mean _____ data

Part 1 · Notes and examples (fill in each note, then work the example)

Which form?: Know the vertex: y = _____. Know the x-intercepts: y = _____.

Example 1

Write an equation of the parabola with vertex (2, 5) that passes through (4, −3).

Example 2

Write an equation of the parabola with x-intercepts −1 and 4 that passes through (2, 12).

Differences: Constant first differences → _____; constant second differences → _____.

Example 3

Tell whether the data are linear, quadratic, or neither. Write a function for the data.

x	0	1	2	3	4	5
y	2	5	12	23	38	57

Part 2 · Partner practice

Partner practice

Write an equation of the parabola with vertex (−3, 1) that passes through (−1, 9).